PRACTICE EXAM

Difficulty: MEDIUM

Questions: 10

Medium Difficulty Exam

Multiple Choice Questions (4 points each)

Instructions: Choose the best answer for each question.

Question 1: According to the text, what principle was the Introduction Lab meant to reintroduce?
A) Ohm's Law
B) Kirchhoff's Laws
C) Superposition Principle
D) Thevenin's Theorem

Question 2: Which PyQt6 module contains the QMainWindow class?
A) QtCore
B) QtGui
C) QtWidgets
D) GStreamer

Question 3: In the experimental setup, what frequency was supplied by the signal generator?
A) 1 kHz
B) 10 kHz
C) 20 kHz
D) 40 kHz

Question 4: Which GStreamer pipeline option is best for dynamic cams?
A) Option A
B) Option B
C) Option C
D) All options are equally optimal

Short Answer Questions (6 points each)

Instructions: Answer each question in 2-3 complete sentences.

Question 5: Briefly describe the objective of the Introduction Lab as stated in the text.

Question 6: What is the purpose of the QApplication class in PyQt6, and how is it typically used in a program?

Question 7: Explain the superposition principle in the context of the experiment described in the document.

Problem-Solving Questions (10 points each)

Instructions: Answer each question with detailed analysis and application of the concepts.

Question 8: Compare and contrast the two options (A & B) for rendering a GStreamer video pipeline inside PyQt. Discuss the trade-offs between performance and flexibility, and provide a potential use case where each option would be more suitable.

Question 9: Based on the provided code snippets, describe how you would implement a grid of video thumbnails (e.g., a 2x4 grid) using PyQt6. Explain the relevant classes and methods involved.

Question 10: Analyze the experimental results presented in the tables. Assuming Table P1 (not provided) has vI(rms) = 2.5 V, discuss possible reasons for any discrepancies between the vI(rms) values in Table P1 and Table E3 and explain how neglecting the 10 ohms of resistance could affect the experimental results.